

R. Bernadett Vlocskó

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PhD-trained organic chemist specializing in small-molecule synthesis, reaction optimization, and route design, seeking to advance drug development by developing efficient, scalable, and selective processes suitable for early- and late-stage development and manufacturing.

- 6+ years of hands-on work in synthetic chemistry labs, focusing on reaction optimization and catalysis for biologically active compounds
- Peer-reviewed publications in high-impact journals including *Journal of Medicinal Chemistry* and *Tetrahedron Green Chemistry*
- Vast experience in organic chemistry teaching labs (10 semesters) and mentoring students (2 undergraduate and 6 graduate)
- 3+ years leading computational chemistry projects using Gaussian, ORCA, Schrödinger suites
- ~1 year of industry experience delivering projects in cross-functional, interdisciplinary chemistry teams

Education

University of Massachusetts Boston, MA

2020 - Expected May 2026

PhD – Organic Chemistry

Bela Torok's Research [Group](#)

Thesis: Computer-aided development of novel antioxidants

Budapest University of Technology and Economics, Hungary

2017- 2019

Master of Science – Chemical Engineering (Drug Development & Manufacturing)

Functional Organic Materials Research [Group](#), HUN-REN Research Centre for Natural Sciences (RCNS)

Thesis: Dynamically controllable interfacial systems containing photoactive azobenzenes

University of Szeged, Hungary

2014 - 2017

Bachelor of Science – Chemistry

MTA-SZTE Stereochemistry Research Group

Thesis: Cinchona alkaloid-catalyzed asymmetric Michael addition of acetylbutyrolactone to activated olefins

Skills

Experimental	Thorough and safe laboratory technique, experience in the handling air and moisture sensitive reagents in multi-step syntheses.
Instrumentation	NMR (¹ H, ¹³ C, ¹⁹ F), UV-Vis, FTIR, GC-QTOF-MS, HPLC-MS, Flash Chrom., XRPD, PLM, TGA, DSC
Computational	Quantum mechanics (QM), molecular mechanics (MM), docking, reactivity and mechanism analysis
Software & Packages	OriginPro, Mestrelab, Gaussian 16, ORCA 6.0, BioLuminate (Schrödinger)

Experience

PhD research

January 2020 – Present

Synthesis

- Developed a scalable, metal- and acid-free protocol for 2-alkyl/aryl benzimidazoles (n = 20) and benzothiazoles (n = 18), achieving quantitative yields under mild conditions.
- Elucidated reaction kinetics and mechanistic pathways using ¹⁹F NMR reaction profiling to support process optimization.
- Developed a microwave-assisted condensation protocol to synthesize a broad library of hydrazone derivatives (n = 34), significantly reducing reaction times and simplifying workup.
- Designed and optimized a high-yielding and selective synthesis of diaryl hydrazone quaternary ammonium salts (n = 12).
- Purified and characterized in-process intermediates and final products using routine chromatographic and spectroscopic techniques to ensure purity and structural confirmation.
- Led laboratory safety operations within the doctoral research group, ensuring compliance with institutional safety protocols.

Computational

- Developed and validated integrated MM/QM and DFT computational workflows to design and optimize small-molecule antioxidants, benchmarking functionals and identifying key physicochemical and structural drivers of activity.
- Advanced a lead diaryl hydrazone candidate from *in silico* design through *in vitro* validation (human trophoblast cells) to patent application, in collaboration with Beth Israel Medical Center.

Master's research

Sept 2017 – July 2019

- Designed and synthesized azobenzene-based molecular switches and immobilized them on polydopamine/gold (PDA/Au) composite surfaces to create robust, reversible photoswitchable interfaces.
- Demonstrated fast, wavelength-selective surface photoisomerization independent of binding group and spacer length, enabling modular and tunable surface functionality.
- Characterized photoresponsive surface behavior using UV-Vis spectroscopy to quantify switching efficiency, reversibility, and kinetic performance.

Agios Pharmaceuticals, Cambridge, MA

June – August 2024

Intern in Drug Substance Group, CMC, R&D

- Identified, synthesized, and fully characterized two novel salt forms of widely used generic drugs (cimetidine and loratadine).
- Validated simulated single-molecule packing arrangements through XRPD pattern similarity analysis.
- Integrated the CCDC Database and Mercury software into the in-house salt/cocrystal screening platform.

Hovione, Lisbon, Portugal

August – December 2019

Intern in Inhalation and Advanced Drug Delivery Group, R&D

- Developed and optimized a dry powder liposomal formulation of fluticasone propionate in DSPC for sustained lung delivery.
- Optimized high-pressure homogenization and spray-drying processes to produce reproducible, controlled particle properties.

Sanofi, Chinoi, Budapest, Hungary

July – August 2018

Intern in Process Safety and Process Control Laboratory, R&D

- Leveraged the CHETAH simulation program for chemical thermodynamic and hazard evaluation.
- Provided group training and authored tutorial materials.

Teaching and mentorship

- Supervised undergraduate (2) and graduate students (6) in experimental design, data interpretation, and laboratory techniques.
- Served as teaching assistant for five years in organic chemistry laboratory courses (188 stud. total).
- Collaborated with multidisciplinary, multicultural teams across chemistry, engineering, and technical operations.
- Demonstrated strong written communication through authoring book chapters, review articles and technical reports.

Presentations

1. **Vlocskó, R. B.**, Mastugin, M., Gustin, L., Zsengellér, Zs., Török, M., Török, B. Exploring Antioxidant Effects of Organofluorine Diphenyl Hydrazone Quaternary Salts in Preeclampsia Therapy: a DFT Approach. *ACS Spring Meeting 2025*.
2. **Vlocskó, R. B.**, Mishra, M., Stoica, A., Gustin, L., Török, B. Catalyst-free Approaches in Green Synthesis: Environmentally Benign Synthesis of Benzimidazoles and Benzothiazoles in a Sustainable Solvent. *28th Annual GC&E Conference 2024*.
3. **Vlocskó, R. B.**, Mastugin, M., Török, B., Török, M. Assessing Antioxidant Activity Through Theory and Experiments: A Comparative DFT Study of Phenol and Thiophenol Analogues. *ACS Spring Meeting 2024*.

Peer-reviewed publications

1. Zsengellér, K. Zs., Mastugin, M., Fusco, R. A., **Vlocskó, R. B.**, Ferguson, C., Pintye, D., Nasim, H., Salahuddin, S., Young, C. B., Török, B., Török, M. Impact of Structural Features on the Antioxidant Activity of Organofluorine Diaryl Hydrazones. *Molecules*, **2026**, 31, 78.
2. **Vlocskó, R. B.**, Mastugin, M., Zsengellér, K. Zs., Török, M., Török, B. Development of Diaryl Hydrazones for Alleviation of Mitochondrial Oxidative Stress in Preeclampsia. *Journal of Medicinal Chemistry*, **2025**, 68, 10075-10091.
3. **Vlocskó, R. B.**, Mastugin, M., Török, B., Török, M. Correlation of Physicochemical Properties with Antioxidant Activity in Phenol and Thiophenol Analogues. *Scientific Reports*, **2025**, 15, 73.
4. **Vlocskó, R. B.**, Mishra, M., Stoica, A., Gustin, L., Török, B. Catalyst-free Synthesis of Substituted Benzimidazoles and Benzothiazoles in a Sustainable Solvent. *Tetrahedron Green Chemistry*, **2024**, 3, 100035.
5. **Vlocskó, R. B.**, Xie, G., Török, B. Green Synthesis of Aromatic Nitrogen-Containing Heterocycles by Catalytic and Non-Traditional Activation Methods. *Molecules* **2023**, 28, 4153.
6. Schäfer, C., Cho, H., **Vlocskó, R. B.**, Xie, G., Török, B. Recent Advances in the Green Synthesis of Heterocycles: From Building Blocks to Biologically Active Compounds. *Current Organic Synthesis*, **2022**, 19, 426-462.
7. Kunfi, A., **Vlocskó, R. B.**, Keresztes, Z., Mohai, M., Bertóti, I., Ábrahám, Á., Kiss, É., London, G. Photoswitchable Macroscopic Solid Surfaces Based on Azobenzene-Functionalized Polydopamine/Gold Nanoparticle Composite Materials: Formation, Isomerization and Ligand Exchange. *ChemPlusChem*, **2020**, 85, 797.

Awards

Vincent T. Mastricola, Jr. Graduate Fellowship	<i>Top performing grad student, UMB Chemistry Department (2025)</i>
CSM Dean's Doctoral Research Fellowship	<i>UMB College of Science & Mathematics (2025)</i>
Ciba Green Chemistry Travel Award	<i>ACS Green Chemistry Institute (2024)</i>
Graduate Teaching Award	<i>UMB Chemistry Department (2021)</i>
Ivan Villax Scholarship	<i>Hovione, Portugal (2019)</i>
New National Excellence Program Scholarship	<i>National Research and Innovation Office of Hungary (2016 – 2017)</i>

Volunteer experience

Official Notetaker	<i>Computational and Medicinal Chemistry School, Cambridge, MA (2025)</i>
Session Presider	<i>Computational Division, ACS Spring Meeting (2025)</i>
STEM Outreach Mentor	<i>UMass Boston–BCLA-McCormack School partnership (2021 – 2022)</i>
Organizer	<i>Eötvös Loránd College for Advanced Studies's student conference and pre-college summer camp (2015, 2017)</i>